Question	Answer	Marks	Partial Marks
6(b)(ii)	$\frac{-11 \pm \sqrt{11^2 - 4 \times 4 \times -300}}{2 \times 4}$	**B2**	**B1** for $\sqrt{11^2 - 4(4)(-300)}$ or better or for $\frac{-11+\sqrt{q}}{2\times 4}$ or $\frac{-11-\sqrt{q}}{2\times 4}$
	–10.14 and 7.39	**B2**	**B1** for each or **SC1** for final answers –10.1 or –10.144 to –10.143 **and** 7.4 or 7.393 to 7.394 or –10.14 **and** 7.39 seen in working or for –7.39 **and** 10.14 as final answer
6(b)(iii)	11.4 or 11.39…	**1**	**FT** *their* positive root + 4
7(a)(i)	13	**1**	
7(a)(ii)	3	**2**	**M1** for $\text{h}\left(\frac{10}{30}\right)$ oe soi or $27^{\frac{10}{x}}$
7(a)(iii)	$\frac{7-x}{2}$ oe final answer	**2**	**M1** for $x = 7 - 2y$ or $y - 7 = -2x$ or $7 - y = 2x$ or $-\frac{y}{2} = -\frac{7}{2} + x$ oe
7(b)	0.75 oe final answer	**3**	**M1** for $\frac{10}{2x+1} = 4$ **M1** for $10 = 8x + 4$ or better
7(c)	$\frac{70-19x}{x(7-2x)}$ or $\frac{70-19x}{7x-2x^2}$ final answer	**3**	**M1** for $x + 10(7 - 2x)$ or better isw **B1** for common denominator $x(7 - 2x)$ oe isw
7(d)	3 final answer	**1**	
8(a)(i)	$\frac{m-7}{5}$ oe final answer	**2**	**M1** for $5p = m - 7$ or $p + \frac{7}{5} = \frac{m}{5}$
8(a)(ii)	$[\pm]\sqrt{\frac{y^2-h}{2}}$ or $[\pm]\sqrt{\frac{h-y^2}{-2}}$ oe final answer	**3**	**M1** for first correct step isolate term in p or divide by ±2 **M1** for second correct step FT *their* first step
8(b)(i)	$\begin{pmatrix} 0 \\ 5 \end{pmatrix}$	**1**	
8(b)(ii)	$\begin{pmatrix} -3 \\ -1 \end{pmatrix}$	**1**	

Question	Answer	Marks	Partial Marks
8(b)(iii)	3.22 or 3.216... to 3.220...	**6**	**B3** for [angle AOB =] 36.8 or 36.9 or 36.84 to 36.87 or **M2** for tan[AOB] = $\frac{3}{4}$ oe or for [AOB =]2 × $\sin^{-1}$ $\left(\frac{\sqrt{(5-4)^2+(0--3)^2}}{10}\right)$ oe or for cos [AOB =] $\frac{5^2+5^2-\left(\sqrt{(5-4)^2+(0--3)^2}\right)^2}{2\times5\times5}$ oe or **M1** for recognition of right-angle with perpendicular from B to OA or x-axis or for [AB^2 =] $(5-4)^2+(0--3)^2$ or better oe or (*their* AB)^2 = $5^2 + 5^2 - 2 \times 5 \times 5 \times \cos OAB$ oe **M2** for $\frac{\textit{their}\text{ angle } AOB}{360}\times2\times\pi\times5$ oe or **M1** for radius = 5 soi
9(a)	171 or 171.0…	**3**	**M2** for $\frac{7.6}{160}\times60\times60$ oe or **M1** for $\frac{7.6}{160}$ or $\frac{7.6}{2\frac{2}{3}}$ or $\frac{7.6}{2\text{ min }40\text{ sec}}$ If 0 scored, **SC1** for answer 189 or 188.6 to 188.7
9(b)(i)	77 [min] 20 [s]	**4**	**M3** for $\frac{32}{12}\times29$ oe or **B2** for 4640 or 1.29 or 1.288 to 1.289, $\frac{58}{45}$ oe or 32 laps or 29 laps or **M2** for $2^5\times 5 \times 29$ oe or **M1** for 2 m 40 sec ÷ (2 m 40 sec – 2 m 25 sec) soi for 2 m 25 sec ÷ (2 m 40 sec – 2 m 25 sec) soi or for an attempt to find LCM or 23 200 seen or correctly find prime factors of 145 or 160 or for $\frac{7.6}{145}$ or $\frac{7.6}{2\frac{5}{12}}$ or $\frac{7.6}{2\text{ min }25\text{ sec}}$ oe, provided SC1 not earned in part (a)
9(b)(ii)	220.4	**2**	**M1** for *their* (b)(i) ÷ 2min 40 sec [× 7.6] oe or *their* (a) × *their* (b)(i) ÷ 60 oe